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**Seventh Semester B.E. Degree Examination, Dec.08/Jan.09**  
**Object Oriented Analysis & Design**

Time: 3 hrs.

Max. Marks:100

**Note: Answer any FIVE full questions.**

- 1
  - a. Why object orientation is necessary? Give reasons. (04 Marks)
  - b. Differentiate between object oriented approach and structured approach. (06 Marks)
  - c. Describe the following terms and give an example for each. (10 Marks)
    - (i) Class hierarchy
    - (ii) Inheritance
    - (iii) Polymorphism
    - (iv) Aggregation
    - (v) Object Identity
- 2
  - a. Using an example of Water-Fall approach, explain how the software development process can be viewed as series of transformation. (06 Marks)
  - b. Describe the quality measures for building high-quality software. (04 Marks)
  - c. With neat diagram, explain the macroprocesses of the object-oriented system development approach. (10 Marks)
- 3
  - a. What are the phases of object modeling technology? Describe each phase briefly. (04 Marks)
  - b. Explain the Booch system development processes. (06 Marks)
  - c. Explain the following methodologies: (i) Use case (ii) OOSE (iii) OOSE (10 Marks)
- 4
  - a. Define pattern. Explain the essential components on reading a pattern. (06 Marks)
  - b. Give the differences between design pattern and frameworks. (04 Marks)
  - c. With neat diagram, explain the processes and components of the Unified approach. (10 Marks)
- 5
  - a. How uses and extends associations are useful in use-case modeling? Describe briefly with an example. (04 Marks)
  - b. Why documentation is an important part of analysis? Explain the guidelines for developing effective documentation. (06 Marks)
  - c. Explain the different use-cases used in Vianet – Bank ATM system. (10 Marks)
- 6
  - a. What are the four main approaches for identifying classes? Explain in detail. (10 Marks)
  - b. Briefly describe the guidelines for identifying super-sub relationship. (04 Marks)
  - c. Define a part-of-relationship with an example. Describe its properties and patterns. (06 Marks)
- 7
  - a. Explain in detail the object oriented design Axioms and Corollaries. (10 Marks)
  - b. Describe briefly the design patterns created by 'Kurotsuchi'. (04 Marks)
  - c. What do you mean by coupling? Explain the different types of coupling by specifying degree of coupling. (06 Marks)
- 8
  - a. Define OODBMS. Give the differences between an OODBMS and traditional DBMS. (04 Marks)
  - b. Explain the characteristics of Multi-database systems. (06 Marks)
  - c. How objects are mapped with relational systems by different mapping capabilities? Explain with an example for each capability. (10 Marks)